

PROLONGED LABOR

Definitⁿ

The labor is said to be prolonged when the combined duratⁿ of the first & second stage is more than the arbitrary time limit of 18 hours.

Labor is considered prolonged when the cervical dilatatⁿ rate is less than 1cm/hr & descent of the presenting part is < 1cm/h for a period of min. 4 hrs obs.

~~Causes~~ It can be due to:-

- Protracted cervical dilatatⁿ in 1st stage
- Inadeq descent of the presenting part during the first or second stage of labor

Causes

First stage

- Fault in power - Abn (1) uterine contractⁿ such as ~~late~~ inertia or incoordinate uterine contractⁿ
- " " passage - Contracted pelvis, cervical dystocia, pelvic tumor or even full bladder
- " " passenger - Malpositⁿ
Malpresentatⁿ
congenital anomalies
- Others - Injudicious admⁿ of sedatives & analgesics

Second stage

- Fault in power - 1) Uterine inertia
2) Inability to bear down
3) Regional anesthesia/analgesia
4) Constrictⁿ ring
- " " passage - 1) CPD
2) Undue resistance
3) Soft tissue pelvic tumor

Fault in passenger - 1) Malposⁿ
2) Malpresentatⁿ
3) Big baby
4) Congenital anomalies

Diagnosis - Partograph

First stage

- When duratⁿ is more than 12 hrs
- Cx dilatⁿ - $< 1 \text{ cm/hr}$ - primi
 $< 1.5 \text{ cm/hr}$ - multi
- Rate of descent is $< 1 \text{ cm/h}$ - primi
 $< 2 \text{ cm/h}$ - multi

- Labor considered abⁿ when
ceruicograph crosses the alert
line & falls on zone 2
Interventⁿ is required when
it crosses the actⁿ line & falls
on zone 3

Note

Actⁿ line is drawn
4 hrs to right of
alert line
An interval of
4 hrs is allowed
to Δ s delay in
active phase
& for
interventⁿ

Disorders of Active phase

Protracted
active
phase

↓
when rate
of cx dilatⁿ
is $< 1.2 \text{ cm/h}$
in primi &
 $< 1.5 \text{ cm/h}$
in multi

Arrest
disorders

↓
no cx dilatⁿ
occurs after 2
hrs in active
phase

No descent for
a period of more
than 2 hrs → arrest of descent

Secondary arrest

active phase commences
(1) but stops or slows
for 2 hrs or more prior
to full dilatⁿ

Second stage

Mean duratⁿ

50 mins - Nullipara

30 mins - Multipara

Diagnosed when duratⁿ exceeds 2 hrs in nullipara
& 1 hr in multipara

Disorders of second stage

Protractⁿ
of descent



when
descent
of PP
is at less
than 1cm/h
in nullipara
or less than
2cm/h in
multipara

Arrest
of descent



no progress
in descent
for at least
2 hrs

Complications

FETAL

- (1) Hypoxia
- (2) Intrauterine infectⁿ, pneumonia
- (3) Intracranial stress or hemorrhage
- (4) ↑ operative delivery

MATERNAL

- (1) Distress
- (2) Chorioamnionitis
- (3) PPH
- (4) Trauma to genital tract
- (5) ↑ operative delivery
- (6) Puerperal sepsis
- (7) Subinvolutⁿ

Treatment

Preventⁿ

- Antenatal or early intranatal detectⁿ
- Use of partograph
- Selective & judicious augmentatⁿ of labor by low ROM & f/b oxytocin drip
- Change of posture in labor
- Emotional support.
- Adeq. analgesia
- Avoid dehydratⁿ

Actual Rx

1) Find cause of prolonged labor

Primⁱ
↓
Inadeq
uterine
activity

Multⁱ
↓
CPD

2) correctⁿ of dehydratⁿ & Ketoacidosis by rapid RL infusⁿ

Definitive Rx

First stage delay

uterine activity suboptimal
↓
Amniotomy
↓
oxytocin infusⁿ
↓
effective pain relief

complicatⁿ
↓
C/S

Second stage delay

FHS reassuring
↓
expectant Mx
↓
vaginal delivery

FHS not reassuring
↓
forup or ventouse C/S

Prolonged Latent phase

Mean duratⁿ

8 hrs - Primiⁿ

4 hrs - Multiⁿ

Latent phase that exceeds 20 hrs in primi or 14 hrs in multi is abnⁿ

Causes

- 1) Unripe cx
- 2) Malposⁿ & Malpresentⁿ
- 3) CPD
- 4) Premature ROM
- 5) IOL
- 6) Early onset of regional anesthetic

• No danger to mother or fetus

Mx

- 1) Expectant Mx
- 2) Rest & analgesic
- 3) when augmentatⁿ is decided → Medical IOL
- 4) Avoid amniotomy
- 5) Not indicatⁿ for c/s